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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 323 — Past Questions & Materials

by Renance

Course: MAR 323

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — MAR 323 Marine Auxiliary Machineries 2020 2021 Second Semester

Level 300 • Second Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING**

EXAMINATION: First Semester

SESSION: 2019/2020

DATE: 24/07/21



COURSE CODE: MAR 323

COURSE TITLE: Marine Auxiliary Machineries

TIME ALLOWED: 2Hrs

INSTRUCTION: ANSWER ANY FOUR QUESTION

1a). 90 percent of the world cargos are transported through the sea. Describe Ro-ro vessel, Reefer vessel, Bulk carrier ship and Container ship with reference to their contribution in transportation of goods from one destination to another. [12.5 marks].

b). What are the responsibilities of a Naval Architecture and Marine Engineer in ship design and construction. [2 marks]

c). A ship is conventionally divided into three distinct areas. List them. [2 marks].

d). Define a ship. [1 mark].

2a). Most ships use diesel engine as one of their prime movers. As a marine engineer, outline five reasons that made you to prefer a diesel engine over other internal combustion engines. [5 marks].

b). Diesel engines as internal combustion engines can be classified into five categories. List them and the make-up of each category. [6.5 marks].

c). Describe four stroke cycle of diesel engine. [4 marks].

d). Outline the difference between medium speed engine and high speed engine. [1 mark]

e). Define Scavenging [1 mark].

3a). Technology advances have made combined power plants utilization possible in modern vessels. Outline two merits of using them. [1.5 marks]

b). Describe the following combined power plants that has made in road in marine and offshore industries.

i). CODAG [4 marks].

ii). COGAS [4 marks].

iii). COSAG [4 marks].

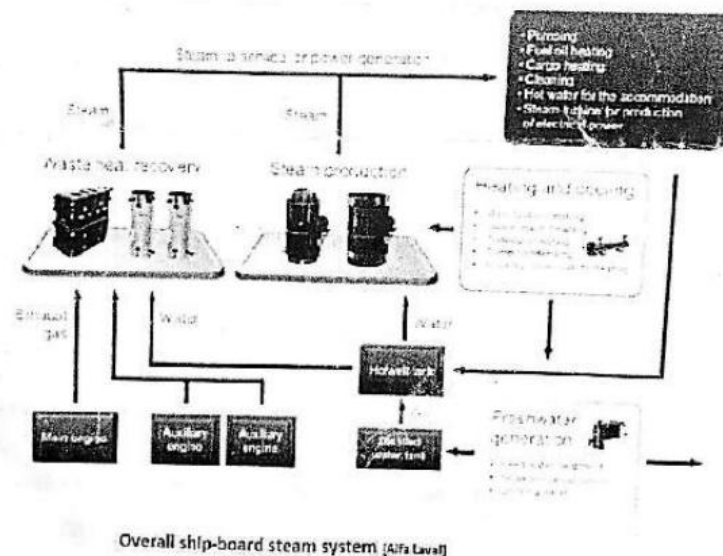
iv). CODAD [4 marks].

4a) Valve is an important component onboard vessel. Define a valve and list functions of five components of it. [6 marks].

b). List five factors that are considered in valve selection for modern ship design and construction. [3.5 marks].

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- c). Valves can be functionally classified into five. List them. Also, what is the difference between electric valve actuator and hydraulic one. [4 marks].
- d). Identify a machine that can be used to raise liquids from a low level to a higher one. State two categories of the machine and their differences. [4 marks].
- 5a). In modern ships, there are various system onboard it. As a marine engineer describe cooling system and bilge and ballast system [10 marks].
- b). Define heat exchanger and list two types of it. Also outline the difference between the two types of heat exchanger. [4 marks].
- c). Outline deck machineries and their functions. Also state a reason for carrying out watch keeping on modern ships. [3.5 marks].
- 6a). Explain the diagram of overall shipboard steam system below. [12 marks].



- b). Outline the similarity between the auxiliary boilers and the exhaust gas economiser. [1.5 marks].
- c). State the a function of the stabilizer, bow thrusters, compressor and a rudder onboard modern ship. [4 marks].

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